

SQL Assessment

Que-1

```
CREATE TABLE Worker (  
  WORKER_ID INT PRIMARY KEY,  
  FIRST_NAME VARCHAR(50),  
  LAST_NAME VARCHAR(50),  
  SALARY INT,  
  JOINING_DATE DATETIME,  
  DEPARTMENT VARCHAR(50)  
);
```

```
INSERT INTO Worker  
(WORKER_ID, FIRST_NAME, LAST_NAME, SALARY, JOINING_DATE, DEPARTMENT)  
VALUES  
(1, 'Monika', 'Arora', 100000, '2014-02-20 09:00:00', 'HR'),  
(2, 'Niharika', 'Verma', 80000, '2014-06-11 09:00:00', 'Admin'),  
(3, 'Vishal', 'Singhal', 300000, '2014-02-20 09:00:00', 'HR'),  
(4, 'Amitabh', 'Singh', 500000, '2014-02-20 09:00:00', 'Admin'),  
(5, 'Vivek', 'Bhati', 500000, '2014-06-11 09:00:00', 'Admin'),  
(6, 'Vipul', 'Diwan', 200000, '2014-06-11 09:00:00', 'Account'),  
(7, 'Satish', 'Kumar', 75000, '2014-01-20 09:00:00', 'Account'),  
(8, 'Geetika', 'Chauhan', 90000, '2014-04-11 09:00:00', 'Admin');
```

1. Print all Worker details ordered by FIRST_NAME Ascending and DEPARTMENT Descending

```
SELECT *  
FROM Worker  
ORDER BY FIRST_NAME ASC, DEPARTMENT DESC;
```

2. Print details for Workers with the first names "Vipul" and "Satish"

```
SELECT *  
FROM Worker  
WHERE FIRST_NAME IN ('Vipul', 'Satish');
```

3. Print details of Workers whose FIRST_NAME ends with 'h' and contains six alphabets

```
SELECT *  
FROM Worker  
WHERE FIRST_NAME LIKE '_____h';
```

4. Print details of Workers whose SALARY lies between 100000 and 300000

```
SELECT *  
FROM Worker  
WHERE SALARY BETWEEN 100000 AND 300000;
```

5. Fetch duplicate records having matching data in some fields of a table

Find duplicate salaries:

```
SELECT SALARY, COUNT(*) AS Total  
FROM Worker  
GROUP BY SALARY  
HAVING COUNT(*) > 1;
```

6. Write an SQL query to show the top 6 records of a table

MySQL

```
SELECT *  
FROM Worker  
LIMIT 6;
```

7. Write an SQL query to fetch the departments that have less than five people in them

```
SELECT DEPARTMENT, COUNT(*) AS Total_Employees  
FROM Worker  
GROUP BY DEPARTMENT  
HAVING COUNT(*) < 5;
```

8. Write an SQL query to show all departments along with the number of people in there

```
SELECT DEPARTMENT, COUNT(*) AS Total_Employees  
FROM Worker  
GROUP BY DEPARTMENT;
```

9. Write an SQL query to print the name of employees having the highest salary in each department

```
SELECT W.FIRST_NAME,  
       W.DEPARTMENT,  
       W.SALARY  
FROM Worker W  
WHERE W.SALARY = (  
    SELECT MAX(SALARY)  
    FROM Worker  
    WHERE DEPARTMENT = W.DEPARTMENT  
);
```

Que-2

```
CREATE TABLE Student (  
    StdID INT PRIMARY KEY,  
    StdName VARCHAR(50),  
    Sex VARCHAR(10),  
    Percentage INT,  
    Class VARCHAR(5),
```

```
Sec CHAR(1),  
Stream VARCHAR(20),  
DOB DATE  
);
```

```
INSERT INTO Student
```

```
(StdID, StdName, Sex, Percentage, Class, Sec, Stream, DOB)
```

```
VALUES
```

```
(1001, 'Surekha Joshi', 'Female', 82, '12', 'A', 'Science', '1998-03-08'),  
(1002, 'Maahi Agarwal', 'Female', 56, '11', 'C', 'Commerce', '2008-11-23'),  
(1003, 'Sanam Verma', 'Male', 59, '11', 'C', 'Commerce', '2006-06-29'),  
(1004, 'Ronit Kumar', 'Male', 63, '11', 'C', 'Commerce', '1997-11-05'),  
(1005, 'Dipesh Pulkit', 'Male', 78, '11', 'B', 'Science', '2003-09-14'),  
(1006, 'Jahanvi Puri', 'Female', 60, '11', 'B', 'Commerce', '2008-11-07'),  
(1007, 'Sanam Kumar', 'Male', 23, '12', 'F', 'Commerce', '1998-03-08'),  
(1008, 'Sahil Saras', 'Male', 56, '11', 'C', 'Commerce', '2008-11-07'),  
(1009, 'Akshra Agarwal', 'Female', 72, '12', 'B', 'Commerce', '1996-10-10'),  
(1010, 'Stuti Mishra', 'Female', 39, '11', 'F', 'Science', '2008-11-23'),  
(1011, 'Harsh Agarwal', 'Male', 42, '11', 'C', 'Science', '1998-03-08'),  
(1012, 'Nikunj Agarwal', 'Male', 49, '12', 'C', 'Commerce', '1998-06-28'),  
(1013, 'Akriti Saxena', 'Female', 89, '12', 'A', 'Science', '2008-11-23'),  
(1014, 'Tani Rastogi', 'Female', 82, '12', 'A', 'Science', '2008-11-23');
```

1. Display all records from Student table

```
SELECT * FROM Student;
```

2. Display student name and date of birth

```
SELECT StdName, DOB  
FROM Student;
```

3. Display students whose percentage is greater than or equal to 80

```
SELECT *  
FROM Student  
WHERE Percentage >= 80;
```

4. Display student name, stream and percentage where percentage is more than 80

```
SELECT StdName, Stream, Percentage  
FROM Student  
WHERE Percentage > 80;
```

5. Display all Science students whose percentage is more than 75

```
SELECT *  
FROM Student  
WHERE Stream = 'Science'  
AND Percentage > 75;
```